EAS5450 Quiz
8:35 – 8:45am
Canvas Quizzes

Venture Capital Method

EAS5450 Class #13

Penn Engineering

1

Authorized & Issued Stock Shares

- **Shares authorized at time of incorporation**
  - Fixed maximum number of shares authorized for future issuance
  - Specified in company's Certificate of Incorporation
  - Held by company until issued
  - Par value of shares: $1/Shr typ.
- **Shares issued to shareholders after incorporation**
  - Drawn from authorized shares and sold to shareholders
  - Shares representing company ownership by shareholders
  - Possessed by the shareholders
  - Purchase price received by company treasury upon issuance

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Venture Capital Method

EAS5450 Class #13

Penn Engineering

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Company Valuation

"Terminal Value after N Years: V_N"

- Price/Earnings Multiple (P/E):

  $V_N = P/E \times E_N$

  where E_N = company net earnings in year N.

- Price/Revenues Multiple (P/R):

  $V_N = P/R \times R_N$

  where R_N = company sales revenues in year N.

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XYZ COMPANY INCOME STATEMENT

(Dollars in thousands except per-share amounts)

		Year Ended June 30,	
		2007	2006
1.	Net Sales	$765,000	$725,000
2.	Cost of Sales	535,000	517,000
GROSS MARGIN		230,000	208,000
3.	Selling expenses	76,500	84,000
4.	General and administrative expenses	20,300	25,500
5.	Depreciation and amortization	28,060	25,000
OPERATING INCOME		105,140	73,500
6.	Dividend and interest income	5,200	10,000
7.	Interest expense	16,250	16,750
INCOME BEFORE TAXES		94,090	66,750
8.	Income taxes	41,400	26,250
9.	Extraordinary income (loss)	0	0
NET INCOME		$52,690	$40,500

$$P/E: \frac{\text{Price per share}}{\text{Earnings per share}}$$

$$P/R: \frac{\text{Price per share}}{\text{Net Sales per share}}$$

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P/E Ratio

P/R Ratio

DELL INC

GICS Sub-Industry Peers Computer Hardware

(10 closest companies based on annual sales)

Peer Companies	Ticker Symbol	Mkt Val	P/E from Ops	P/Sales	P/B	Div/Yld	ROE	12MM Sales	1 Year Sales Growth	Revenue Share
SORT	SORT	SORT	SORT	SORT	SORT	SORT	SORT	SORT	SORT	SORT
APPLE COMPUTER INC	AAPL	63026.733	37.5	3.6	7.3	0.000	19.9	17306.000	56.0	4.10
DELL INC	DELL	53010	15.1	.9	15.7	0.000	100.8	56738.000	11.1	13.44
DIEBOLD INC	DBD	2860.899	36.3	1	2.7	1.950	5.0	2783.036	12.6	0.66
FUJITSU LTD	FJTSY	17591.814		0.4	2.2	0.604	6.1	43717.545	-4.6	10.36
GATEWAY INC	GTW	684.342		0.2	2.9	0.000	-11.0	4140.302	13.3	0.98
HEWLETT-PACKARD CO	HPQ	103201.92	18.4	1.1	2.8	0.848	13.5	90016.000	5.7	21.33
INTL BUSINESS MACHINES CORP	IBM	126525.694	14.3	1.4	3.8	1.203	25.2	88505.000	-8.0	20.97
NCR CORP	NCR	6971.938	20.8	1.2	3.3	0.000	23.1	6029.000	-0.4	1.43
NEC CORP	NIPNY	10728.94	42.8	.3	1.4	0.461	3.4	41076.000	-8.1	9.73
SUN MICROSYSTEMS INC	SUNW	17409.91		1.3	2.7	0.000	-13.6	13068.000	18.0	3.10
TOSHIBA CORP	TOSBF	19453.581	46.5	.4	2.6	0.000	5.8	54689.849	-0.3	12.96
Computer Hardware - 500	CMP.I1	366696.078	@NC	1.403	4.113	0.589	16.908		10.499	3.365
S&P 500 Comp-Ltd	I0003	12164054.977	@NC	1.532	2.881	1.745	15.291		12.559	

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EBITDA

Earnings before interest, taxes, depreciation, and amortization are commonly shortened to EBITDA. EBITDA reports a company's profits before interest on debt and taxes owed or paid to the government are subtracted. EBITDA is used to compare the profitability of a company with other companies of the same size in the same industry who may have different levels of debt or different tax situations.

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XYZ COMPANY INCOME STATEMENT

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1. Net Sales	$765,000	$725,000
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6. Dividend and interest income	5,200	10,000
7. Interest expense	16,250	16,750
INCOME BEFORE TAXES	94,090	66,750
8. Income taxes	41,400	26,250
9. Extraordinary income (loss)	0	0
NET INCOME	$52,690	$40,500
10. Earnings per common share before extraordinary item(s)	$3.54	$2.72
11. Earnings per common share: extraordinary item(s)	0.00	0.00
NET INCOME PER COMMON SHARE	$3.54	$2.72

Earnings		$52,690	$40,500
Interest		16,250	16,750
Taxes		41,400	26,250
Depreciation & Amortization		28,060	25,000
EBITDA		$138,400	$108,500
X10 =	**Valuation:**	**$1,384,000**	**$1,085,000**

Assuming a valuation multiplier of 10 times EBITDA

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EXAMPLE: Value of a company in year 5, given income statement data

Sales projected in 5 years	$30MM
COGS	-18MM
Gross Margin	12MM
Sales, General & Administrative expenses and R&D expenses	-7.5MM
Taxes	-2.0MM
Net Income (earnings)	**$2.5MM**

Referring to financial references for comparable companies, we find a P/E = 15.
∴ Value of the company in 5 years: $V_5 = 15 \times \$2.5MM = \$37.5MM$

$V_5 = 15 \times \$2.5MM = \$37.5MM$

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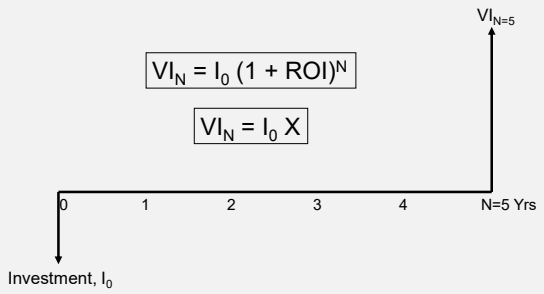
Investment Timeline

Value of Investment after N Years, VI_N



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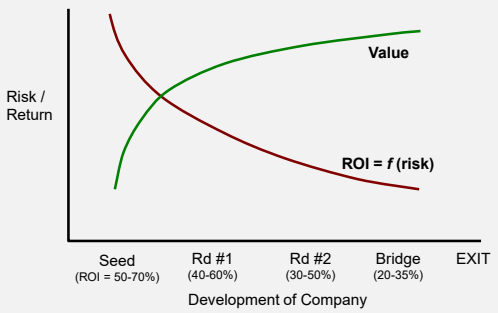
Investment Risk/Return



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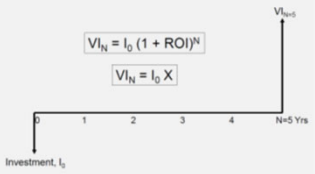

EXAMPLE: Value of a $500,000 investment in year 5, given desired ROI of 50% per annum =

$$VI_5 = \$0.5MM \times (1 + 0.50)^5 = \$3.8MM$$

Alternatively, the investor may simply specify that he/she wants to receive "X" times their investment after N years, in which case:

$$VI_N = I_0 \times X$$

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Terminal Ownership

- % of company investor must own at the time of the liquidity event to achieve target ROI:

$$\%_{\text{Ownership}} = \frac{VI_N}{V_N}$$

where N = years from investment to liquidity event.

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$$\%_{\text{Ownership}} = \frac{VI_N}{V_N}$$

EXAMPLE: Percent ownership required

In 5 years, at the liquidity event, the VC must own:

$$\%_5 = \frac{3.8}{37.5} \times 100\% = 10.1\%$$ of the company.

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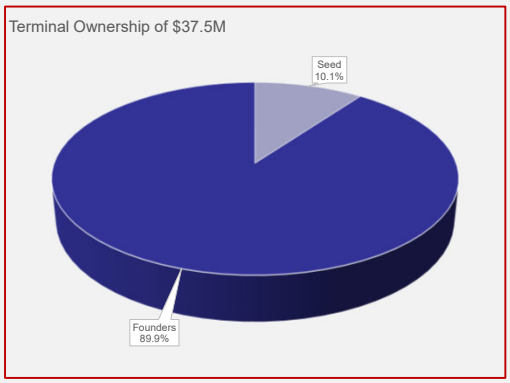

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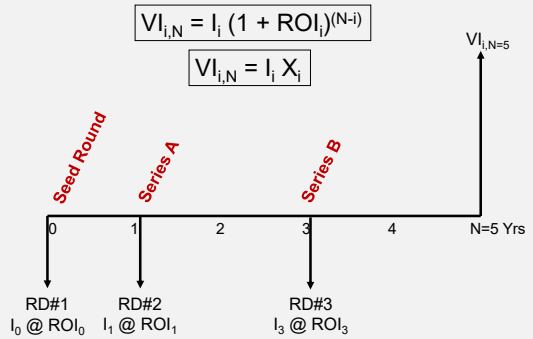
Multiple Investment Rounds

$$VI_{i,N} = I_i\,(1 + ROI_i)^{(N-i)}$$

$$VI_{i,N} = I_i\,X_i$$



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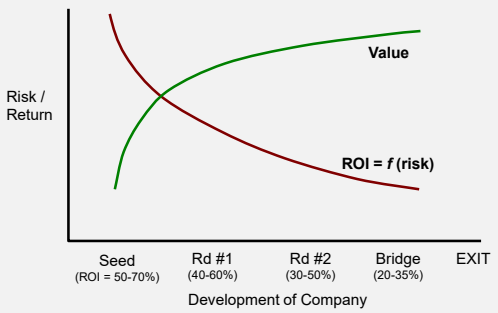
Investment Risk/Return



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Terminal Ownership Revisited

- % of company each investor must own at the time of the liquidity event to achieve target ROI:

$$\%_{Ownership} = \frac{VI_{i,N}}{V_N}$$

Round	Investment	Target ROI	Terminal Ownership
#1	I_0	ROI_0	$[I_0 \times (1 + ROI_0)^5] / V_5$
#2	I_1	ROI_1	$[I_1 \times (1 + ROI_1)^4] / V_5$
#3	I_3	ROI_3	$[I_3 \times (1 + ROI_3)^2] / V_5$

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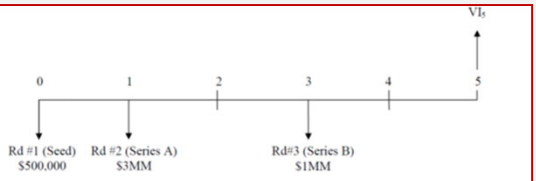

Percent Ownership

Let's assume that due to the gradual reduction of risk in the business, the required ROI for each round is progressively decreased:

Round	Invested	Target ROI	VI_N	Terminal Ownership*
#1 (Seed)	$0.5MM	50%	$3.80MM	10.1%
#2 (Series A)	$3.0MM	40%	$11.50MM	30.7%
#3 (Series B)	$1.0MM	25%	$1.56MM	4.2%

* Calculated based on formulas above.

NOTE: This would leave 55% for the founders, options, or other shares for future negotiations.

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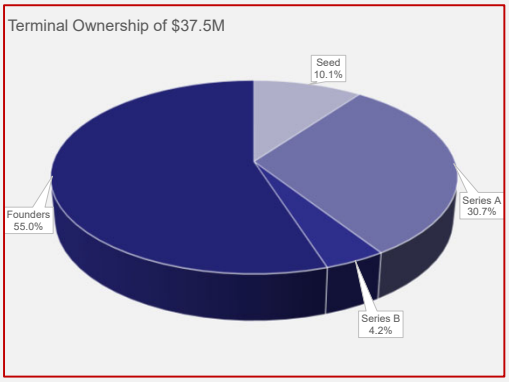

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Retention

$$\text{Retention \%} = 100\% - \sum\left(\%_{\text{Ownership}} \text{ of Later Rounds}\right)$$

Example: Retention percentage for 3 rounds

Round	Retention %
#1 (Seed)	100% - 30.7% - 4.2% = 65.1%
#2 (Series A)	100% - 4.2% = 95.8%
#3 (Series B)	100% = 100%

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% Acquired at Time of Investment

- % of company investor must acquire at the time of investment to achieve target ROI:

$$\%_{\text{Acquired}} = \frac{\left(\%_{\text{Ownership (Terminal)}}\right)}{\left(\text{Retention}\% \div 100\%\right)}$$

∴At the time of their respective investments, each investor must acquire:

Round	% Acquired
#1 (Seed)	10.1% ÷ 0.651 = 15.5%
#2 (Series A)	30.7% ÷ 0.958 = 32.0%
#3 (Series B)	4.2% ÷ 1.0 = 4.2%

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Shares Purchased

New Shares Issued to Investor

$$\%_{\text{Acquired}} = \frac{\text{\# Shares Purchased}}{\text{Total \# Shares Issued}}$$

$$\%_{\text{Acquired}} = \frac{\text{\# Shares Purchased}}{\text{\# Old Shares}^{*} + \text{\# Shares Purchased}}$$

$$\text{\# Shares Purchased} = \frac{\%_{\text{Acquired}}}{\left(1 - \%_{\text{Acquired}}\right)} \times \left(\text{\# Old Shares}\right)$$

* # Old Shares = Number of shares issued pre-investment

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$$\text{\# New Shares Purchased} = \frac{\%_{\text{Acquired}}}{1 - \%_{\text{Acquired}}} \times \text{\# Old Shares}$$

If 1,000,000 shares have been issued (e.g., Founders' shares) before the initial Seed round:

Example: New shares issued at each round of investment

Round	# New Shares Purchased
#1 (Seed)	$\frac{0.155}{1-0.155} \times 1{,}000{,}000 = 183{,}430$ **shares**
#2 (Series A)	$\frac{0.32}{1-0.32} \times (1{,}000{,}000 + 183{,}430) = 556{,}910$ **shares**
#3 (Series B)	$\frac{0.042}{1-0.042} \times (1{,}000{,}000 + 183{,}430 + 556{,}910) = 76{,}300$ **shares**

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Price per Share

$$\text{Price per Share} = \frac{\text{Investment (\$)}}{\text{\# Shares Purchased}}$$

Example: Price per share at each round of investment

Round	Price Per Share
#1 (Seed)	= \$500,000 ÷ 183,430 = **\$2.73**
#2 (Series A)	= \$3,000,000 ÷ 556,910 = **\$5.39**
#3 (Series B)	= \$1,000,000 ÷ 76,300 = **\$13.11**

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Post-Money Valuation

$$\text{Post-Money Valuation} = \frac{\text{Investment (\$)}}{\%_{\text{Acquired}}}$$

or… Post – Money Valuation =
(Price per Share) × (Total # Shares Issued)

Example: Post-Money Valuations

Round	Investment	% Acquired	Price/Shr	Total # Shares Post-Money	Post-Money Valuation
#1 (Seed)	\$500,000	15.5%	\$2.73	1,183,430	\$3.23 MM
#2 (Series A)	\$3,000,000	32.0%	\$5.39	1,740,340	\$9.38 MM
#3 (Series B)	\$1,000,000	4.2%	\$13.11	1,816,640	\$23.81 MM

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Pre-Money Valuation

Pre – Money Valuation =
(Post – Money Valuation) – (Investment)

Or… Pre – Money Valuation =
(Price per Share) × (# Old Shares)

EXAMPLE: Pre-Money Valuations

Round	Post-Money Valuation	Investment	Price/Shr	Total # Shares Pre-Money	Pre-Money Valuation
#1 (Seed)	\$3.23 MM	\$0.5MM	\$2.73	1,000,000	\$2.73 MM
#2 (Series A)	\$9.38 MM	\$3MM	\$5.39	1,183,430	\$6.38 MM
#3 (Series B)	\$23.81 MM	\$1MM	\$13.11	1,740,340	\$22.81 MM

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Terminal Price per Share

$$\text{Price per Share} = \frac{\text{Company Value, } V_N}{\text{Total \# Shares Issued}}$$

Example: Final price per share

$$\frac{\$37.5\text{MM}}{(1{,}000{,}000 + 183{,}430 + 556{,}910 + 76{,}300)\text{ shares}} = \frac{\$37.5\text{MM}}{1{,}816{,}640\text{ shares}} = \$20.64/\text{share}$$

Founders' shares | Seed investor's shares | Series A shares | Series B shares

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Return on Investment

$$\text{ROI} = \left(\frac{\text{Price per share}_{\text{sold}}}{\text{Price per share}_{\text{purchased}}}\right)^{1/N} - 1$$

Example: Returns on investment for each round

Round	ROI
#1 (Seed)	$(\$20.64 \div \$2.73)^{1/5}$-1 = **50% p.a.**
#2 (Series A)	$(\$20.64 \div \$5.39)^{1/4}$-1 = **40% p.a.**
#3 (Series B)	$(\$20.64 \div \$13.11)^{1/2}$-1 = **25% p.a.**

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Convertible Preferred Shares

Non-Compounding Dividend = (Investment) x i x N

Compounding Dividend = (Investment) x $[(1+i)^N - 1]$

where: i = dividend interest rate p.a.
N = years from investment to conversion

$$\text{Additional Common Shares Acquired at Conversion} = \frac{\text{Dividend}}{\text{Price / Share}}$$

$$\text{Distribution} = (\text{Value of Company}) \times \frac{(\text{Total Shares Owned*})}{(\text{Total Shares Issued*})}$$

* Including add'l shares acquired by conversion

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In the above example, suppose the investor in Round #2 negotiated for Convertible Preferred Shares having a non-cash, cumulative, compounding dividend of 5% per annum. As shown above, this investor invests $3,000,000 four years prior to the planned liquidity event and receives 556,910 shares at a price of $5.39 per share. At the time of the liquidity event, the cumulative, compounding dividend would be worth:

$$\text{Dividend} = I\,[(1+i)^N - 1\,] = \$3{,}000{,}000\,[\,1.05^4 - 1\,] = \$646{,}520$$

At the original share price of $5.39, this would convert to:

$$\text{Additional Shares} = \frac{\$646{,}520}{\$5.39/\text{share}} = 119{,}950 \text{ shares}$$

... assuming the shareholder elects to convert the dividend.

In this case the Round #2 investor's total holdings at the liquidity event would be:

Total shares owned = 556,910 + 119,950 = 676,860 shares

The Round #2 investor's distribution from the liquidity event would then be:

$$\text{Distribution} = (\text{Value of Company}) \times \frac{(\text{Total shares owned})}{(\text{Total \# shares issued})}$$

$$= \$37{,}500{,}000 \times \frac{676{,}860}{1{,}816{,}640 + 119{,}950} = \$13{,}125{,}000$$

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Return on investment (ROI) = [Distribution / Investment]$^{1/N}$ – 1
= [$13,125,000 / $3,000,000]$^{1/4}$ – 1
= 44.6% p.a. which, as expected, exceeds their 40% target return.

Note: If the dividend had been cumulative, non-compounding, at the time of the liquidity event the dividend would have been worth:

$$\text{Dividend} = I \times i \times N = \$3{,}000{,}000 \times 0.05 \times 4 = \$600{,}000$$

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Participating Preferred Shares

$$\text{Distribution} = (\text{Investment}) + \left\{ \left[(\text{Value of Company}) - (\text{Investment}) \right] \times \left[\frac{\text{Total Shares Owned}}{\text{Total Shares Issued}} \right] \right\}$$

Example: Distribution on Participating Preferred Stock

In the above example, if only the Round #2 investor has Participating Preferred Stock, then the investor's distribution at the time of the liquidity event would be:

$$\text{Distribution} = I + \left([(\text{Value of Company}) - I] \times \left[\frac{(\text{Total shares owned})}{(\text{Total \# shares issued})} \right] \right)$$

$$= \$3{,}000{,}000 + \left([\$37{,}500{,}000 - 3{,}000{,}000] \times \left[\frac{676{,}860}{1{,}936{,}590} \right] \right)$$

$$= \$15{,}060{,}000$$

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Return on investment (ROI) = [Distribution / Investment]$^{1/N}$ – 1
= [\$15,060,000 / \$3,000,000]$^{1/4}$ – 1
= 49.7% p.a. which, as expected, exceeds their 40% target return.

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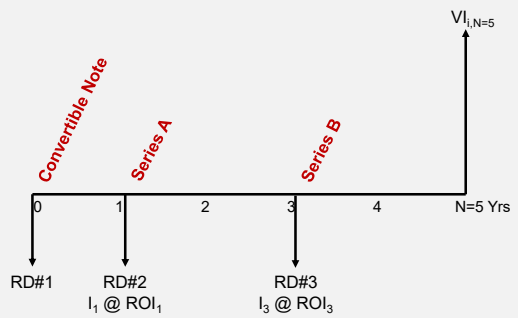
Multiple Investment Rounds

with a Convertible Note



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Convertible Note

- Principal Amount of Note
- Interest Rate (6-8% compounded p.a., typ.)
- Duration (18 months, typ.)
- Minimum Investment of Next Round Triggering Conversion
- Price/Shr Discount When Converting (None or 20% typ.)
- Valuation Cap on Conversion
- Right to Purchase Next Round of Financing
- If Acquisition Prior to Next Round, Premium Over Principal
- Board Seat(s) Not Uncommon
- Consequences if Duration Exceeded

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Number & Price of Converted Shares

- Post-Money Valuation of the Note =
  (Pre-Money Valuation of Series A) x (1-Discount)
- $(\%\ \text{Acquired})_{Note} = \dfrac{(\text{Principal+Accrued Interest on Note})}{(\text{Post−Money Valuation of Note})}$
- Principal + Accrued Interest on Note (PI_{Note}) =
  (Note Principal) x $(1+\text{Note Interest Rate})^N$

  where: N is time between note issuance and Series A conversion.
- # Shares Acquired (S_{Note}) = $\dfrac{\%_{Acquired}}{1-\%_{Acquired}}$ x (# Old Shares)
- Price per share (P_{Note}) = $\dfrac{PI_{Note}}{S_{Note}}$

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Pre-Money Valuation Cap

Recall:
Pre-Money Valuation = (Price per Share) x (# Old Shares)

- Pre-money valuation cap normally refers to the pre-money valuation of the Convertible Note (not to the pre-money valuation of Series A).
- Maximum capped price of Convertible Note shares is:

$$P_{Note,Max} = \frac{\text{Pre−Money Valuation Cap}}{\text{Pre−Money Shares (i.e. "Old Shares")}}$$

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Example: Convertible Note Seed Funding

Assume the Round #1 seed financing in the prior example is a convertible note subject to the following terms:

- Principal: $500,000
- Interest: 6% p.a. compounded
- Term: 18 months
- Minimum round to trigger conversion: $1,000,000
- Price discount on conversion: 20%
- Pre-money valuation cap: $6,000,000

As before, the Series A round occurs one year after the seed round, which is within the note's 18-month expiration term. The Series A investment of $3,000,000 exceeds the $1,000,000 minimum therefore this will trigger conversion of the note. The principal and interest on the convertible note at the time of the Series A round is then:

$$PI_N = \$500{,}000 \times (1.06)^1 = \$530{,}000$$

The pre-money valuation of the Series A round is $6,355,000 as found before. The post-money valuation of the note is then:

$$(\text{Post-Money Valuation})_{Note} = \$6{,}355{,}000 \times (1 - 0.20) = \$5{,}084{,}000$$

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The percent of the company acquired by the convertible note investor at the time of conversion is then:

$$(\%_{Acquired})_{Note} = \frac{(\$530{,}000)}{(\$5{,}084{,}000)} = 10.43\%$$

The number of shares (S_{Note}) acquired by the note's investor at the time of conversion is then:

$$S_{Note} = \frac{0.1043}{1 - 0.1043} \text{ x } (1{,}000{,}000) = 116{,}385 \text{ shares}$$

And finally, the price per share (P_{Note}) acquired by the convertible note investor is:

$$P_{Note} = \frac{\$530{,}000}{116{,}385 \text{ shrs}} = \$4.55/\text{share}$$

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Pre-Money Valuation Cap

Recall: Pre-Money Valuation = (Price per Share) x (# Old Shares)

$$Conversion\ Price\ Cap = \frac{Pre - Money\ Valuation\ Cap}{Pre - Money\ Shares}$$

$$P_{Note\ Cap} = \frac{\$6{,}000{,}000}{1{,}000{,}000} = \$6/\text{share}$$

so the $4.55/share price of the convertible note shares is valid.

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Series A & B Shares

$$\text{\# Shares Purchased} = \frac{\%_{Acquired}}{(1 - \%_{Acquired})} \times (\text{\# Old Shares})$$

Round	# New Shares Purchased	Price/Share
#1 (Seed)	116,385 shares by conversion	$4.55
#2 (Series A)	$\frac{0.3207}{1 - 0.3207} \times (1{,}000{,}000 + 116{,}385) = 527{,}025$ shares	$\frac{\$3{,}000{,}000}{527{,}025} = \$5.69$
#3 (Series B)	$\frac{0.0417}{1 - 0.0417} \times (1{,}000{,}000 + 116{,}385 + 527{,}025) =$ 71,453 shares	$\frac{\$1{,}000{,}000}{71{,}453} = \$14.00$

Note: The $5.69 Series A price found here is consistent with the convertible note price of $4.55 which is discounted 20% from the Series A price.

* # Old Shares = Number of shares issued pre-investment

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Terminal Price per Share

$$\text{Price per Share} = \frac{V_N}{\text{Total \# Shares Issued}}$$

$$\frac{\$37.5MM}{(1{,}000{,}000 + 116{,}385 + 527{,}025 + 71{,}453)shares} = \frac{\$37.5MM}{1{,}714{,}863\ shares} = \frac{\$21.87}{share}$$

Round	ROI
#2 (Series A)	$(\$21.87 \div \$5.69)^{1/4}$-1 = **40% p.a.**
#3 (Series B)	$(\$21.87 \div \$14.00)^{1/2}$-1 = **25% p.a.**

The results above do, in fact, match the desired returns sought by our Series A and Series B investors.

$$\left(\frac{Terminal\ Distribution\ to\ Seed\ Investor}{Seed\ Investment}\right)^{1/5} - 1 = \left(\frac{\$21.87 \times 116{,}385\ shrs}{\$500{,}000}\right)^{1/5} - 1$$

$$= 38.5\%\ p.a.$$

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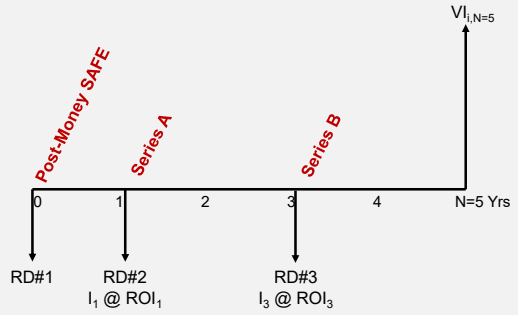
Multiple Investment Rounds

with a Post-Money SAFE



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Post-Money SAFE

- Principal Amount of SAFE Investment
- Minimum Investment of Next Round Triggering Conversion
- Price/Shr Discount When Converting (None or 20% typ.)
- Post-Money Valuation Cap on Conversion
- Right to Purchase Next Round of Financing
- If Acquisition Prior to Next Round, Premium Over Principal
- Board Seat(s) Not Uncommon

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Number & Price of Converted Shares

- Post-Money Valuation of the SAFE =
  (Pre-Money Valuation of Series A) x (1-Discount)

- $(\%\ \text{Acquired})_{SAFE} = \dfrac{(\text{Principal Amount of SAFE})}{(\text{Post-Money Valuation of SAFE})}$

  where the Post-Money Valuation of the SAFE in the denominator is the lesser of the Post-Money Valuation calculated above or the Post-Money Valuation Cap.

- # Shares Acquired $(S_{SAFE}) = \dfrac{\%_{Acquired}}{1-\%_{Acquired}}$ x (# Old Shares)

- Price per share $(P_{SAFE}) = = \dfrac{(\text{Principal Amount of SAFE})}{(S_{SAFE})}$

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Example: Post-Money SAFE Seed Funding

Assume the Round #1 seed financing in the prior example is a Post-Money SAFE subject to the following terms:

- Principal: $500,000
- Minimum round to trigger conversion: $1,000,000
- Price discount on conversion: 20%
- Post-Money Valuation Cap: $6,500,000

As before, the Series A round occurs one year after the seed round, which is within the SAFE's 18-month expiration term. The Series A investment of $3,000,000 exceeds the $1,000,000 minimum therefore this will trigger conversion of the SAFE. The pre-money valuation of the Series A round is $6,355,000 as found before. The post-money valuation of the SAFE is then:

$$(\text{Post-Money Valuation})_{SAFE} = \$6{,}355{,}000 \text{ x } (1 - 0.20) = \$5{,}084{,}000$$

This value is less than the Post-Money Valuation Cap of the SAFE and, therefore, will be used in the calculations that follow.

The percent of the company acquired by the SAFE's investor at the time of conversion is then:

$$(\%_{Acquired})_{SAFE} = \frac{(\$500{,}000)}{(\$5{,}084{,}000)} = 9.84\%$$

The number of shares (S_{SAFE}) acquired by the note's investor at the time of conversion is then:

$$S_{SAFE} = \frac{0.0984}{1-0.0984} \text{ x } (1{,}000{,}000) = 109{,}078 \text{ shares}$$

And finally, the price per share (P_{SAFE}) acquired by the SAFE's investor is:

$$P_{SAFE} = \frac{\$500{,}000}{109{,}078 \text{ shrs}} = \$4.58/\text{share}$$

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Series A & B Shares

$$\text{\# Shares Purchased} = \frac{\%_{Acquired}}{\left(1-\%_{Acquired}\right)} \times \left(\text{\# Old Shares}\right)$$

Round	# New Shares Purchased	Price/Share
#1 (Seed)	109,078 shares by conversion	$4.58
#2 (Series A)	$\frac{0.3207}{1-0.3207} \times (1{,}000{,}000 + 109{,}078) = 523{,}576$ shares	$\frac{\$3{,}000{,}000}{523{,}576} = \$5.73$
#3 (Series B)	$\frac{0.0417}{1-0.0417} \times (1{,}000{,}000 + 109{,}078 + 523{,}576) =$ 70,985 shares	$\frac{\$1{,}000{,}000}{70{,}985} = \$14.09$

Note: The $5.73 Series A price found here is consistent with the SAFE shares price of $4.58 which is discounted 20% from the Series A price.

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Terminal Price per Share

$$\text{Price per Share} = \frac{V_N}{\text{Total \# Shares Issued}}$$

$$\frac{\$37.5MM}{(1{,}000{,}000 + 109{,}078 + 523{,}576 + 70{,}985)shares} = \frac{\$37.5MM}{1{,}703{,}639\ shares} = \frac{\$22.01}{share}$$

As before, to check our work, we should confirm that the return on investment for each round does in fact equal our original targeted returns.

Round	ROI
#2 (Series A)	$(\$22.01 \div \$5.73)^{1/4}$-1 = **40% p.a.**
#3 (Series B)	$(\$22.01 \div \$14.09)^{1/2}$-1 = **25% p.a.**

The results above do, in fact, match the desired returns sought by our Series A and Series B investors.

$$\left(\frac{Terminal\ Distribution\ to\ Seed\ Investor}{Seed\ Investment}\right)^{1/5} - 1 = \left(\frac{\$22.01 \times 109{,}078\ shrs}{\$500{,}000}\right)^{1/5} - 1$$

$$= 36.9\%\ p.a.$$

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